

CSCI-338-Computer Science Theory
Fall 2025
HW 02

Instructor: Adiesha Liyanage

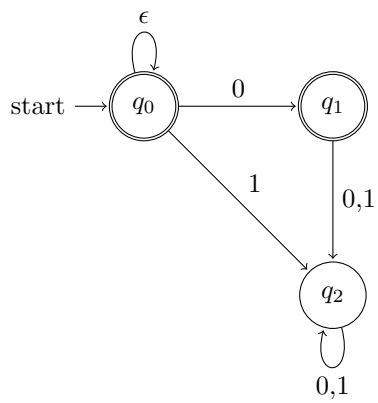
This assignment is **due on September 23, 2025**. You will need to use Latex to generate a single pdf file and upload it under Assignment 02 on Canvas. There will be a penalty of not using Latex (to finish the assignment). This is not a group-assignment, so you must finish the assignment by yourself. **Figures can be hand drawn and inserted to the pdf.**

1 Problem 01

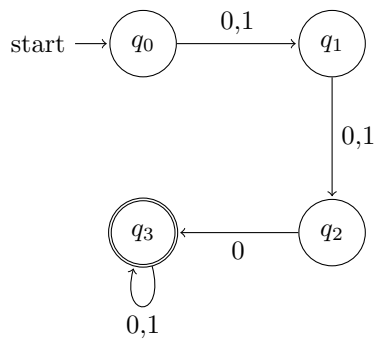
Give state diagrams of DFAs (finite automaton) that recognizes the following languages. Consider the alphabet is $\Sigma = \{0, 1\}$.

1. $A = \{\epsilon, 0\}$
2. $B = \{w : w \text{ has length at least 3 and its third symbol is a } 0\}$
3. $C = \{w \text{ starts with } 0 \text{ and has odd length, or starts with } 1 \text{ and has even length}\}$

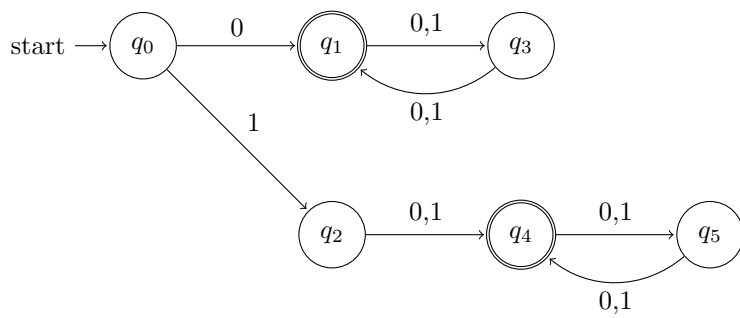
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2.



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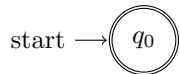


2 Problem 02

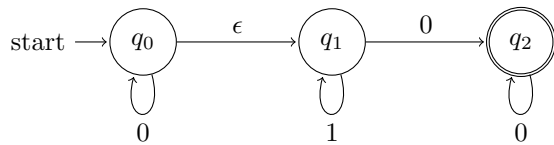
Give state diagrams of NFAs with the specified number of states recognizing each of the following languages. In all parts, the alphabet is $\Sigma = \{0, 1\}$

1. The language $\{\epsilon\}$.
2. The language $0^*1^*0^+$ with three states.
3. The language $A = \{w : w \text{ contains an even number of 0s, or contains exactly two 1s}\}$ with 6 states.

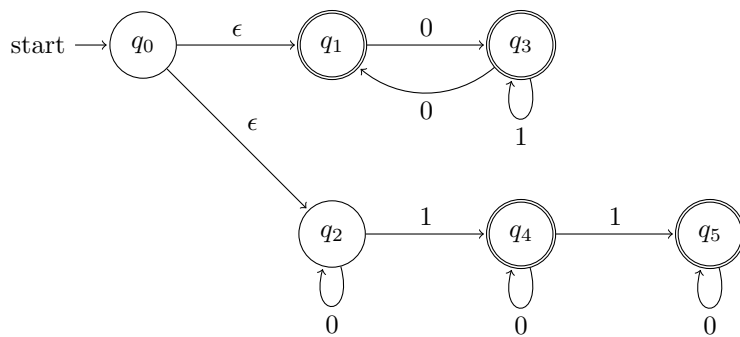
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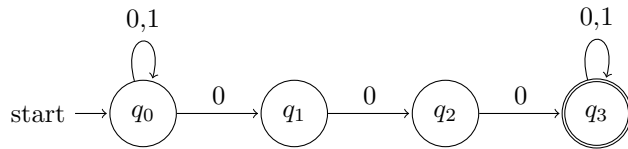


3 Problem 03

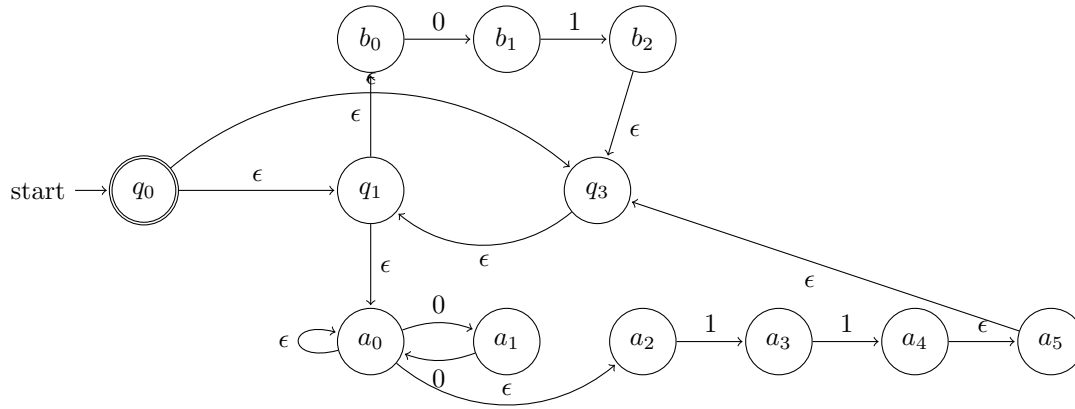
Use the procedure described in Lemma 1.55 to convert the following regular expressions to nondeterministic finite automata.

- $(0 \cup 1)^*000(0 \cup 1)^*$
- $((00)^*(11) \cup 01)^*$
- $\emptyset^*$

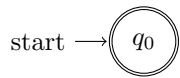
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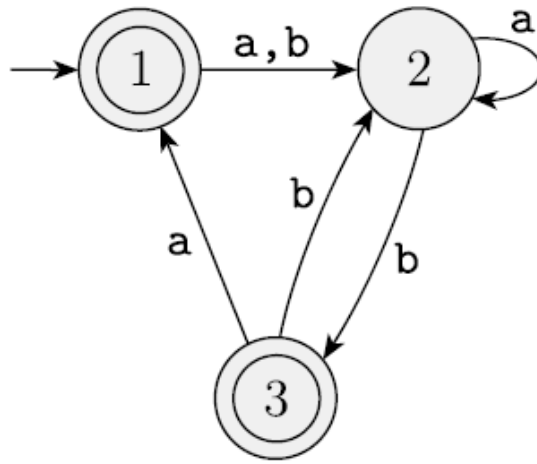
3.



4 Problem 04

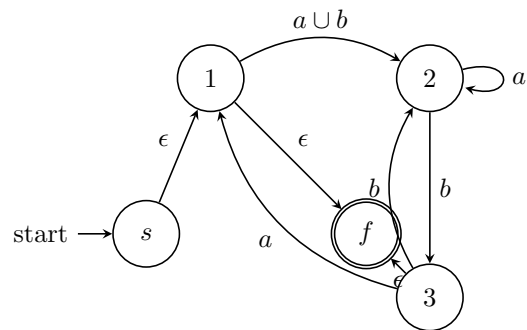
Use the procedure described in Lemma 1.60 to convert the following finite automata to regular expressions.

Problem: Convert the automaton to a regular expression using the GNFA construction (Lemma 1.60).

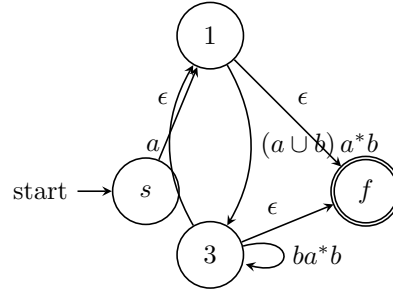


(b)

Figure 1: Finite automaton

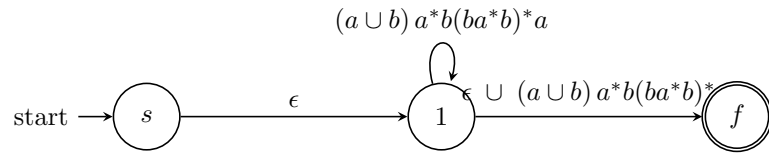


Rip 2



$$1 \rightarrow 2 = (a \cup b) a^* b, \quad 2 \rightarrow 3 = ba^* b.$$

Rip 3



$$1 \rightarrow 1 = (a \cup b) a^* b (ba^* b)^* a, \quad 1 \rightarrow f = \epsilon \cup (a \cup b) a^* b (ba^* b)^*.$$

Rip 1

Final answer:

$$((a \cup b) a^* b (ba^* b)^* a)^* (\epsilon \cup (a \cup b) a^* b (ba^* b)^*).$$

5 Problem 05

For each of the following languages, give two strings that are members and two strings that are not members—a total of four strings for each part. Assume the alphabet $\Sigma = \{a, b\}$ in all parts.

1. a^*b^*
2. $aba \cup bab$
3. $(\epsilon \cup a)b$
4. $(a \cup ba \cup bb)\Sigma^*$
 1. Members: "aabbb", "aa" Non-members: "ba", "abab"
 2. Members: "aba", "bab" Non-members: "aa", "aab"
 3. Members: "b", "ab" Non-members: "a", "aa"
 4. Members: "a", "baa" Non-members: "b", "ab"